

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**First Semester of M. Tech. Examination (IT)****Dec 2012****IT702: High Performance Computer Networking****Date: 19.12.2012, Wednesday****Time: 10:00 am To 01:00 pm****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Write the answer of following Questions. [11]**

- (a) What is use of CLP in ATM? [1]
- (b) Explain the following terms: [2]
 1. External Virtual Circuit
 2. Internal Virtual Circuit
- (c) Give the appropriate reason for the statement: "Failure and Congestion can influence routing decision" [2]
- (d) What are the advantages of Frame Relay over X.25? [2]
- (e) Justify: "X.25 is more reliable than Frame Relay Network" [2]
- (f) Explain following terms [2]
 1. Committed Burst Size
 2. Excess Burst Size

Q - 2 Write the answer of following Questions. [12]

- (a) What are the five characteristics of Virtual Path Connection? [5]
- (b) How does Frame Relay manage Traffic Rate Management? [7]

OR

- (b) What are the service provided by AAL (AAL1, AAL2, AAL3/4, AAL5) layer of ATM? [7]

Q - 3 Write the answer of following Questions. (Attempt any TWO) [12]

- (a) What are the data link control functions provided by LAPF core Protocol?
- (b) What are the two basic method of congestion control in packet switching network? How do they differ with each other?
- (c) How does ATM achieve Constant Bit Rate for Audio and Video signals?

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M.Tech. (EC) Examination

December- 2012

EC – 701 Advanced High Speed Networks

Date: 21.12.2012, Friday

Time: 10.00 a.m to 1.00 p.m

Maximum Marks: 70

Instructions: (1) Attempt all questions.

(2) Answer to the two sections must be written in separate answer books.

(3) Figures to the right indicate full marks

(4) Make suitable assumptions and draw neat figures whenever required.

SECTION -I

- Q-1(a) Give three differences between X.25 and Frame Relay. 03
- (b) What do you mean by internetworking? Discuss the function of router. 04
- Q-2(a) With neat sketch discuss the basic architecture of ATM for an interface between user and network. 07
- (b) Give the characteristics of some high speed LANs. Explain Gigabit Ethernet in detail. 07

OR

- Q-2 (a) Discuss Fiber Channel Protocol Architecture in Brief. 05
- (b) Give the difference between datagram circuit switching and virtual circuit approach for packet switching techniques. 05
- (c) Suppose that an 11- Mbps 802.11b LAN is transmitting 64-byte frames back-to-back over a radio channel with bit error rate of 10^{-7} . How many frames per second will be damaged on average? 04
- Q-3 (a) Explain the Data Encryption Standard (DES) in detail. 07
- (b) What is the need of protocol architecture? Explain the architecture of frame relay. 07

OR

- Q-3(a) Describe transposition cipher for computer network security 05
- (b) Write a short note on ALL services 05
- (c) Explain the operation of TCP/IP reference model with figure. 04